

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Department of Mining Engineering

2. **Subject Name:** Underground Metal Mining **L-T-P:** 3-0-0 **Credits:** 3

3. **Pre-requisites:** Development of mineral deposits (MI20201)

4. **Syllabus and reference books:**

Syllabus:

- Metalliferous ore deposits, Ore to metal - process examples;
- Underground horizontal, inclined and vertical development for stoping preparation; Raising and Shaft sinking;
- Stopping blocks and level interval;
- Methods of stoping: Classification of methods, open stoping, supported stoping – Breast Stopping, Stope and Pillar, Room and Pillar, Shrinkage Stopping, Cut and Fill stoping, Sub-level stoping;
- Underground ore movement network;
- Stope supports: Props, Pillars, Rock bolts, Cable Bolts;
- Caving methods: Sub-level caving, Block caving; Cavability assessment; High productivity methods: Long hole stoping, Vertical Crater Retreat ;
- Qualitative and numerical approaches in selection of stoping methods;
- Backfilling: Types and properties of Fill materials, Mill tailings, Paste fill, Filling preparation, hydraulic transportation and safety in filling operation.

Reference Books:

1. Hartman H. L., Mutmansky J. M. (2007). Introductory Mining Technology, 2nd Edition, Wiley.
2. Darling P. (Ed.) (2011). SME handbook Vol 1 and 2, 3rd Edition, SME publications.
3. Hustrulid W. A., Bullock R. L., Bullock R. C. (Ed.) (2001). Underground Mining Methods: Engineering Fundamentals and International Case Studies, SME Publications.
4. Potvin Y., Thomas E. D., Fourie A. (Ed.) (2005). Handbook on Mine Fill, Australian Centre for Geomechanics, The University of Western Australia.
5. Bateman, A. M., & Jensen, M. L. (1950). Economic mineral deposits (Vol. 259). New York: Wiley.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Metalliferous ore deposits, Ore to metal - process examples	2
2.	Underground horizontal, inclined and vertical development for stoping preparation; Raising and Shaft sinking	6
3.	Stoping blocks and level interval	2
4.	Methods of stoping: Classification of methods, open stoping, supported stoping – Breast Stoping, Stope and Pillar, Room and Pillar, Shrinkage Stoping, Cut and Fill stoping, Sub-level stoping	12
5.	Underground ore movement network	2
6.	Stope supports: Props, Pillars, Rock bolts, Cable Bolts	2
7.	Caving methods: Sub-level caving, Block caving; Cavability assessment; High productivity methods: Long hole stoping, Vertical Crater Retreat	5
8.	Qualitative and numerical approaches in selection of stoping methods	2
9.	Backfilling: Types and properties of Fill materials, Mill tailings, Paste fill, Filling preparation, hydraulic transportation and safety in filling operation	3
Total number of hours		36